

Notes: Farboodi, Singal, Veldkamp, and Venkateswaran (RFS, 2025)

Valuing Financial Data

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1 Summary

Farboodi, Singal, Veldkamp, and Venkateswaran ask a practical but difficult question: **how much should an investor be willing to pay for a stream of financial data?** The answer is not simply the data vendor's price. It is the investor-specific private value of the data, measured by how much the data improve the investor's portfolio decision after accounting for the investor's wealth, risk tolerance, investment mandate, existing information set, and market liquidity.

The paper develops a sufficient-statistics approach in a noisy rational expectations setting. The central simplification is that an investor does not need to estimate every other market participant's preferences, endowments, information, and trading constraints. Instead, the relevant equilibrium information is summarized by return moments: expected returns, unconditional return variance, and conditional return variance with and without the data. In other words, a data set is valuable to an investor when it lowers the forecast-error variance of returns that the investor can trade on.

The empirical exercises value two broad types of data. First, the paper values I/B/E/S median earnings-growth forecasts for investors with different wealth levels, different trading mandates, different existing data sets, and different trading horizons. Second, the paper values macroeconomic information, such as advance information about realized GDP growth. Across these applications, the main pattern is strong heterogeneity: the same data can be worth very little to one investor and much more to another.

The most important economic results are threefold. First, investor wealth and risk tolerance can make the same data worth orders of magnitude more in dollar terms. Second, price impact compresses these differences sharply because large investors move prices against themselves when they trade on their information. Third, investment style matters: data about earnings forecasts are more valuable for investors trading growth, large-cap, S&P 500, or multiasset portfolios than for investors restricted to small or value portfolios. Macroeconomic information, by contrast, has a broader value across investment styles.

The paper's broader contribution is to link information valuation to data-market demand. If investors have highly dispersed private values for the same data, then aggregate demand for data can be inelastic: a change in the subscription price moves only a small mass of investors across the buy-versus-not-buy threshold. At the same time, market illiquidity reduces high-end valuations and can make data demand more elastic by flattening the distribution of private values.

2 Big picture

- **Research question.** How can an investor assign a dollar value to a financial data stream?
- **Core object.** The private value of data is the certainty-equivalent utility gain from having the data, converted into dollars.
- **Methodological contribution.** The value can be computed from sufficient statistics based on equilibrium return moments, rather than from a full structural estimate of every investor's characteristics.
- **Central sufficient statistics.** Expected returns, unconditional return variances, and conditional return variances with and without the data.
- **Economic insight.** Data are more valuable when they reduce forecast errors for assets the

investor can and wants to trade.

- **Main empirical implication.** Heterogeneous investors value the same data very differently; price impact reduces and compresses those valuations.

3 The economic problem

Financial data are useful because they help investors forecast asset payoffs, returns, or sentiment. But the value of a data set is not universal. The same signal can be highly valuable for one investor and nearly useless for another.

The investor-specific value depends on several dimensions:

- **Wealth and risk tolerance.** A wealthier investor can scale up trades and therefore monetize a useful signal more, but also faces larger price impact.
- **Investment style.** An investor constrained to trade only small stocks, value stocks, growth stocks, or the S&P 500 can use only data that help predict returns in that investable universe.
- **Existing data.** A new data set is less valuable when its information is already spanned by the investor’s current database.
- **Trading horizon.** A signal may be more or less useful depending on whether the investor trades quarterly, annually, or at another frequency.
- **Market liquidity.** If trades have price impact, then trading on the signal erodes the payoff from the information.

The paper separates **private value** from **market price**. A data subscription price is an equilibrium transaction price. The private value is how much a particular investor would be willing to pay, given how the data improve that investor’s decision problem.

4 Model intuition

The model is a noisy rational expectations environment with heterogeneous investors. Investors receive information, choose portfolios, and earn profits from asset returns. A data source changes the investor’s posterior beliefs about future asset returns or payoffs. The investor then trades based on those beliefs.

The key object is the investor’s profit vector, denoted here by Π_{it} , for the assets that investor i can trade. The paper shows that the value of an information set can be expressed through the conditional variance of this profit vector. Under perfect competition, a simplified version of the utility expression is:

$$\tilde{U}_i(I_{it}) = \frac{1}{2} \mathbb{E}[\Pi_{it}]' \mathbb{V}[\Pi_{it} | I_{it}]^{-1} \mathbb{E}[\Pi_{it}] + \frac{1}{2} \text{Tr} \left(\mathbb{V}[\Pi_{it}] \mathbb{V}[\Pi_{it} | I_{it}]^{-1} - I \right) + r \rho_i \bar{w}_{it}.$$

This expression gives the paper’s main intuition. Better data reduce $\mathbb{V}[\Pi_{it} | I_{it}]$, the conditional variance or forecast-error variance of the investor’s tradable returns. A lower forecast-error variance lets the investor take better positions. The gain is larger when expected returns are high, when the

data reduce uncertainty substantially, and when the investor has enough risk tolerance and wealth to trade on the information.

With price impact, the investor's information is still valuable, but the effective conditional variance is adjusted for the fact that trades move prices. The paper's price-impact results show that even modest illiquidity can substantially reduce the value of data, especially for large or low-absolute-risk-aversion investors.

5 Dollar value of data

The paper converts utility gains into dollars using a certainty-equivalent logic. If X_t is the data stream being valued and Z_t is the investor's existing information set, the dollar value is:

$$\text{Value}_i(X_t \mid Z_t) = \frac{1}{\rho_i} \left[\tilde{U}_i(Z_t \cup X_t) - \tilde{U}_i(Z_t) \right].$$

This formula says that data are worth the amount of risk-free wealth that would make the investor indifferent between having the data and not having it. The division by ρ_i translates utils into dollars. Because ρ_i is local absolute risk aversion, wealth matters: for the same relative risk aversion, richer investors have lower absolute risk aversion and can value the same useful signal much more in dollar terms.

6 Sufficient-statistics implementation

The empirical implementation is simple. Let R_t be the vector of excess returns in the investor's investable universe. Let X_t be the data stream being valued and let Z_t be the investor's already-owned data. The paper estimates two forecasting systems:

$$R_t = \beta_1 X_t + \beta_2 Z_t + \varepsilon_t^{XZ}, \quad R_t = \gamma_2 Z_t + \varepsilon_t^Z.$$

The residual covariance matrix from the first regression estimates $\mathbb{V}[R_t \mid X_t, Z_t]$. The residual covariance matrix from the second estimates $\mathbb{V}[R_t \mid Z_t]$. The value of the new data comes from the utility difference implied by replacing the larger forecast-error variance with the smaller one.

This is the core sufficient-statistics result. The analyst needs:

- the unconditional expected return vector $\mathbb{E}[R_t]$,
- the unconditional return covariance matrix $\mathbb{V}[R_t]$,
- the conditional covariance with current data, $\mathbb{V}[R_t \mid Z_t]$,
- the conditional covariance with current plus new data, $\mathbb{V}[R_t \mid X_t, Z_t]$,
- the investor's risk tolerance, investment mandate, and price-impact environment.

The paper's important claim is that these objects summarize all the relevant information for valuing the data in the investor's decision problem.

7 Empirical exercise 1: I/B/E/S forecast data

The first application values I/B/E/S median earnings-growth forecast data. The data are used to predict quarterly excess returns on constructed portfolios: Small, Large, Growth, Value, and S&P 500. The baseline control information includes the consumption-wealth ratio *cay*, and in the multiasset case the value-weighted mean price of the S&P 500 portfolio.

7.1 Wealth and price impact

Table 1 shows the role of wealth and liquidity. Under perfect competition, the I/B/E/S data are worth about \$3,130 to an investor with \$500,000 in wealth, but about \$1.06 million to an investor with \$250 million in wealth. This is the wealth-scaling effect: the same data are more valuable when the investor can trade on them at larger scale.

When the investor faces price impact, the value for the \$250 million investor falls from about \$1.06 million to about \$129,000. The high-wealth investor loses more from price impact because larger information-based trades push prices against the investor. This is one of the paper’s most important quantitative messages: **liquidity is a central determinant of financial-data value.**

7.2 Investment style

Table 2 shows that investment mandates strongly affect data value. I/B/E/S earnings-growth forecasts are most valuable for investors trading Large, Growth, S&P 500, or multiasset portfolios. They have little value for an investor who only trades the Small portfolio in the reported specification. This is because the data reduce return forecast errors for some portfolios much more than for others.

The implication is that the same data vendor product can have very different private values across clients. This can make data markets segmented and can help explain low price elasticity of aggregate data demand.

7.3 Already-owned data

Table 3 asks whether I/B/E/S forecasts are substitutes for other data already held by the investor. The estimated values vary substantially when the investor’s existing data set changes. The value is especially high when the investor’s only existing data are the Baker-Wurgler sentiment index, and lower when the investor already has all the listed control series.

The important lesson is that data value is marginal. A data set should be priced not by asking whether it is informative in isolation, but by asking whether it adds information beyond the investor’s current database.

7.4 Trading horizon

Table 4 compares quarterly and annual trading horizons. In the reported exercise, annual-horizon estimates are larger than the quarterly baseline. The paper treats trading horizon as another dimension of heterogeneity, but the contrast is less central than the wealth, style, and price-impact dimensions.

8 Empirical exercise 2: macroeconomic information

The second main application values macroeconomic information. Table 5 computes the value of a hypothetical data source that allows the investor to perfectly forecast realized quarterly GDP growth in advance.

This data source is valuable across a broader set of investment styles than the I/B/E/S forecast data. For example, macroeconomic information is valuable even for investors restricted to Small portfolios, because GDP growth helps predict returns for that investment style. The multiasset investor with \$250 million in wealth under perfect competition would be willing to pay about \$1.37 million for the data in the reported exercise; with price impact, the value falls to about \$836,000.

The contrast between Tables 2 and 5 is useful: **firm-forecast data** have highly style-specific value, while **macro data** can have broader cross-style value.

9 Equilibrium information and price learning

Appendix G and Figure H1 emphasize that prices themselves are information. If prices partially reveal what other investors know, then the value of a private data series depends on how much information is already embedded in prices. Table G1 shows that learning from prices can change data values materially. Table G2 shows how alternative price-impact calibrations change data values. Table G3 reports the return moments that underlie the sufficient-statistics calculations.

Figure H1 shows a subtle point: conditioning on price information can change the value of a data source in a nonmonotonic way, depending on the correlation between the price signal and the data signal. This is why the sufficient-statistics approach is useful. It lets the researcher measure the net value of the data after price informativeness and other investors' information have already been reflected in equilibrium returns.

10 Identification and measurement logic

The paper is not an event-study identification paper. Its empirical contribution is measurement. The identifying logic is that if a data set is useful for an investor's portfolio choice, it should reduce return forecast errors relative to the investor's current information set. The regression residual covariance matrices estimate these forecast errors.

The strength of the approach is its portability. The same recipe can value public data, private data, data about one asset, data about many assets, data about dividends, and data about sentiment. The limitation is that the estimated value is only as reliable as the forecasting specification, the timing of the data, the return panel, and the price-impact assumptions.

11 What to remember

Table 1: Memory map

Element	Meaning
Question	How much should an investor pay for a stream of financial data?
Object	Investor-specific private value, not the observed subscription price.
Mechanism	Data reduce forecast-error variance for returns the investor can trade.
Sufficient statistics	Expected returns, unconditional variances, and conditional variances with and without the data.
Dollar conversion	Utility gain divided by local absolute risk aversion.
Heterogeneity	Wealth, risk tolerance, investment style, existing data, horizon, and price impact all change data value.
Main empirical lesson	The same data can be worth almost nothing to one investor and a large amount to another.
Liquidity lesson	Price impact sharply lowers and compresses data valuations, especially for large investors.
Data-market implication	Highly dispersed values imply low price elasticity of aggregate data demand.

12 Strengths and limitations

12.1 Strengths

- The method is directly connected to portfolio choice and therefore gives an economically interpretable dollar value.
- The sufficient-statistics approach avoids estimating a full equilibrium model of all other investors.
- The framework accommodates investor heterogeneity, price informativeness, and price impact.
- The empirical examples show why data vendors may face very different willingness-to-pay across clients.
- The paper provides a bridge between asset-pricing information models and the economics of data markets.

12.2 Limitations

- The dollar values depend on the return-prediction specification and the assumption that the data timing is usable before trading decisions.
- The method values marginal data relative to a specified existing information set; a different baseline database can change the answer.

- Price impact is difficult to measure precisely, and alternative estimates can materially change data values.
- The framework estimates private values, not equilibrium subscription prices or negotiated data contracts.
- The approach is most transparent in linear-normal settings; nonlinear signals may require transformations or richer Bayesian estimation.

13 Extracted figures and tables from the paper

Table 1
Risk tolerance, liquidity, and data value

	Perfect competition	With price impact
<i>A. Investor with \$500,000 wealth.</i>		
Utility gain	0.082 (0.02)	0.053 (0.02)
Expected profit	0.008 (0.01)	0.007 (0.01)
Variance reduction	0.075 (0.013)	0.046 (0.013)
Dollar value (in \$000)	3.13 (0.75)	2.03 (0.75)
<i>B. Investor with \$250m wealth.</i>		
Utility gain	0.082 (0.02)	0.01 (0.005)
Expected profit	0.008 (0.01)	0.001 (0.001)
Variance reduction	0.075 (0.013)	0.009 (0.005)
Dollar value (in \$000)	1,062.69 (254.07)	128.96 (66.08)
Time periods	140	140

The table reports the valuation of the quarterly value-weighted means of I/B/E/S median forecast earnings growth for Growth and S&P 500 portfolios. Quarterly data between 1985 and 2019. The dependent variables in (13) and (14) are the vector of returns, in excess of a 3-month Treasury (CMT), for five portfolios: {Small, Large, Growth, Value, S&P 500}. The specification includes a constant and control variables (cay and value-weighted mean price of the S&P 500 portfolio). The case with price impact assumes Kyle's lambda $\lambda = 1.5 \times 10^{-8}$. Absolute risk aversion is from (9). All values are annualized, and dollar values are reported in thousands of USD. Standard errors are calculated using a wild bootstrap.

Table 2
Investment styles and data value

	Investment style					
	Small	Large	Growth	Value	S&P500	Multiasset
<i>A. Perfect competition.</i>						
Dollar value (in \$000) for						
Investor with \$500,000 wealth	0.0 (-)	0.33 (0.24)	0.34 (0.28)	0.15 (0.25)	1.0 (0.46)	3.13 (0.75)
Dollar value (in \$000) for	0.0	110.9	116.42	52.27	340.22	1062.69
Investor with \$250m wealth	(-)	(81.11)	(96.31)	(86.14)	(157.62)	(254.07)
<i>B. With price impact.</i>						
Dollar value (in \$000) for						
Investor with \$500,000 wealth	0.0 (-)	0.33 (0.24)	0.34 (0.28)	0.15 (0.25)	1.0 (0.46)	2.03 (0.75)
Dollar value (in \$000) for	0.0	53.86	58.48	11.33	56.35	128.96
Investor with \$250m wealth	(-)	(38.21)	(46.18)	(18.83)	(30.15)	(66.08)
Time periods	140	140	140	140	140	140

Value of I/B/E/S data (quarterly value-weighted means of I/B/E/S median forecasted earnings growth for Growth and S&P 500 portfolios). Quarterly data between 1985 and 2019. Dependent variables in (13) and (14) are returns, in excess of a 3-month Treasury (CMT), for five portfolios: {Small, Large, Growth, Value, S&P 500}. All specifications include a constant and control variables (cay and value-weighted mean price of the corresponding portfolio). In the last column, we control for a constant, cay and the value-weighted mean price of the S&P 500 portfolio. The price impact panel assumes Kyle's lambda $\lambda = 1.5 \times 10^{-8}$. All values are annualized, and dollar values are reported in thousands of USD. Standard errors are calculated using a wild bootstrap.

(a) Table 1: Wealth, risk tolerance, price impact, and I/B/E/S data value.

(b) Table 2: Investment styles and the value of I/B/E/S forecast data.

Figure 1: Extracted tables showing baseline valuation heterogeneity across wealth, liquidity, and investment style.

Table 3
Previously purchased data

Other data	Dollar value (in \$000) for investor with	
	Wealth: \$500,000	Wealth: \$250m
cay (Baseline)	2.03 (0.75)	128.96 (66.08)
No other data	2.72 (0.72)	119.73 (66.54)
Real CMT-1yr	2.09 (0.75)	142.26 (66.47)
S&P 500 D/P ratio	1.65 (0.7)	117.78 (58.86)
BW Sentiment Index	3.74 (0.86)	234.61 (75.97)
All	1.51 (0.8)	113.48 (63.71)
Time Periods	140	140

Values in each row represent the additional value of I/B/E/S data (specifically, the quarterly value-weighted means of I/B/E/S median forecasted earnings growth for Growth and S&P 500 portfolios) *on top of* the value-weighted mean price of the S&P 500 portfolio, and control variable(s) listed in the first column, with price impact assuming Kyle's lambda $\lambda = 1.5 \times 10^{-8}$. Quarterly data between 1985 and 2019. Dependent variables in (13) and (14) are returns, in excess of a 3-month Treasury (CMT) for five portfolios: {Small, Large, Growth, Value, S&P 500}. All specifications include a constant. All values are annualized, and dollar values are reported in thousands of USD. Standard errors are calculated using a wild bootstrap.

(a) Table 3: How existing data holdings change the marginal value of I/B/E/S forecasts.

Table 4
Trading horizon

	Quarterly	Annual
Dollar value (in \$000) for Investor with \$500,000 wealth	2.03 (0.75)	20.45 (4.86)
Dollar value (in \$000) for Investor with \$250m wealth	128.96 (66.08)	198.73 (184.07)
Time periods	140	35

Data between 1985 and 2019. Dependent variables in (13) and (14) are returns, in excess of a 3-month Treasury (CMT), for five portfolios: {Small, Large, Growth, Value, S&P 500}. All specifications include a constant, cay, and price of S&P 500 portfolio. Data variables being valued are the value-weighted means of I/B/E/S median forecasted annual earnings growth, reported quarterly, for Growth and S&P 500 portfolios. Data variables correspond to earnings forecasts with corresponding time horizon. Values are calculated with price impact assuming Kyle's lambda $\lambda = 1.5 \times 10^{-8}$. All values are annualized, and dollar values are reported in thousands of USD. Standard errors are calculated using a wild bootstrap.

(b) Table 4: Quarterly versus annual trading-horizon valuations.

Figure 2: Extracted tables on investor information sets and trading horizons.

Table 5
Macroeconomic information

	Investment style					
	Small	Large	Growth	Value	S&P 500	Multiasset
<i>A. Perfect competition.</i>						
Dollar value (in \$000) for investor with \$500,000 wealth	2.45 (0.4)	3.29 (0.42)	3.29 (0.44)	1.74 (0.4)	3.84 (0.41)	4.05 (0.99)
Dollar value (in \$000) for investor with \$250m wealth	830.01 (136.73)	1116.13 (142.78)	1117.42 (149.14)	589.57 (134.55)	1303.35 (139.17)	1374.21 (334.93)
<i>B. With price impact.</i>						
Dollar value (in \$000) for investor with \$500,000 wealth	2.45 (0.4)	3.29 (0.42)	3.29 (0.44)	1.74 (0.4)	3.84 (0.41)	4.04 (0.99)
Dollar value (in \$000) for investor with \$250m wealth	276.59 (50.51)	504.29 (65.08)	522.44 (63.84)	126.19 (31.57)	201.83 (26.15)	836.43 (98.24)
Time periods	140	140	140	140	140	140

Quarterly data between 1985 and 2019. Dependent variables in (13) and (14) are returns, in excess of a 3-month Treasury (CMT), for five portfolios: {Small, Large, Growth, Value, S&P 500}. All specifications include a constant and controls for cay, value-weighted mean price of the corresponding portfolio, the realized real GDP growth in the previous quarter and the median forecasted quarterly growth rate in real GDP from the Survey of Professional Forecasters. In the last column the price control is for S&P 500 portfolio. Data variables (X_t in (13)) are the second release estimates of real quarterly GDP numbers as reported by the BEA, expressed as growth rates over the previous quarter. Numbers reported in each column represent the additional value of ex post real GDP growth data (8) on top of the control variables for an investor trading at the quarterly horizon. The case with price impact assumes Kyle's lambda $\lambda = 1.5 \times 10^{-8}$. All values are annualized, and dollar values are reported in thousands of USD. Standard errors are calculated using a wild bootstrap.

(a) Table 5: Value of macroeconomic information across investment styles.

Table G1
Price information and data value

(in \$ 000)	Investment Style					
	Small	Large	Growth	Value	S&P500	Multi-Asset
Value (Data Controls + Price Info)	0.0	53.86	58.48	11.33	56.35	128.96
Value (Data Controls)	0.0	49.63	54.4	8.98	47.67	102.5
Percent Change	–	8.51%	7.49%	26.26%	18.22%	25.82%

Value of I/B/E/S earnings growth forecasts of Growth and S&P 500 portfolios. Quarterly data between 1985 and 2019. Dependent variables in (13) and (14) are returns, in excess of a 3-month Treasury (CMT), for five portfolios: {Small, Large, Growth, Value, S&P 500}. All specifications include a constant and control variables (cay). In the first row, the value-weighted mean price of the corresponding portfolio (value-weighted mean price of the S&P 500 portfolio for the last column) is supplied as an additional control variable, while this control variable is missing in the second row. Data variables being valued are the quarterly value-weighted means of I/B/E/S median forecasted earnings growth for Growth and S&P 500 portfolios. Values are calculated with price impact assuming Kyle's lambda $\lambda = 1.5 \times 10^{-8}$. All values are annualized, and dollar values are reported in thousands of USD. Standard errors are calculated using a wild bootstrap.

(b) Table G1: Price information and data value.

Figure 3: Extracted table on macro information and appendix table on learning from prices.

Table G2
Price impact and data value

(in \$ 000)	Investment Style					
	Small	Large	Growth	Value	S&P500	Multi-Asset
Perfect Competition	0.0	110.9	116.42	52.27	340.22	1062.69
GK Price Impact	0.0	87.77	93.61	25.28	136.0	202.24
Baseline Price Impact	0.0	53.86	58.48	11.33	56.35	128.96

Value of I/B/E/S earnings growth forecasts of Growth and S&P 500 portfolios. Quarterly data between 1985 and 2019. The dependent variables in (13) and (14) are returns, in excess of a 3-month Treasury (CMT), for five portfolios: {Small, Large, Growth, Value, S&P 500}. All specifications include a constant and control variables (cay). In the first row, the value-weighted mean price of the corresponding portfolio (value-weighted mean price of the S&P 500 portfolio for the last column) is supplied as an additional control variable, while this control variable is missing in the second row. Data variables being valued are the quarterly value-weighted means of I/B/E/S median forecasted earnings growth for Growth and S&P 500 portfolios. All values are annualized, and dollar values are reported in thousands of USD. Standard errors are calculated using a wild bootstrap.

(a) Table G2: Alternative price-impact assumptions and data value.

Table G3
Return moments

	Investment Style					
	Small	Large	Growth	Value	S&P500	Multi-Asset
$E[R] \times 100$	11.72	9.75	9.93	6.1	9.49	–
$V[R] \times 10,000$	568.15	889.16	941.92	291.71	260.5	–
$V[R \text{Controls}] \times 10,000$	558.9	870.92	921.61	288.48	256.37	–
$V[R \text{Data} + \text{Controls}] \times 10,000$	560.66	857.9	907.14	286.45	246.87	–
Utility Gain	0.0	0.009	0.009	0.004	0.026	0.082

Quarterly data between 1985 and 2019. Dependent variables in (13) and (14) are returns, in excess of a 3-month Treasury (CMT), for five portfolios: {Small, Large, Growth, Value, S&P 500}. All specifications include a constant and control variables (cay and value-weighted mean price of the corresponding portfolio). In the last column, we control for a constant, cay and the value-weighted mean price of the S&P 500 portfolio. Data variables being valued are the quarterly value-weighted means of I/B/E/S median forecasted earnings growth for Growth and S&P 500 portfolios. The return moments reported are annualized and scaled (as indicated).

(b) Table G3: Return moments behind the sufficient-statistics calculation.

Figure 4: Extracted appendix tables on price impact and return moments.

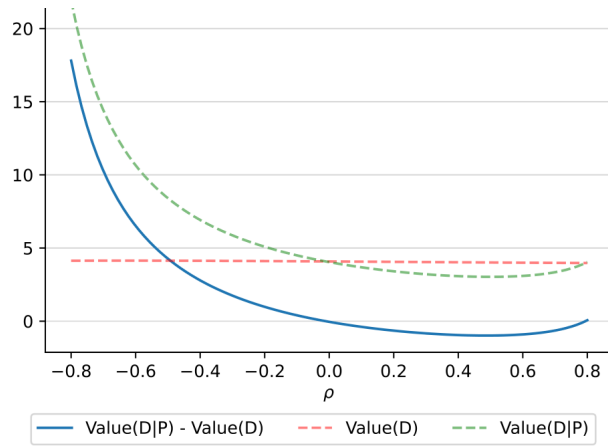


Figure H1
Learning from prices and data value
 Values for a randomly generated data series is plotted, with and without the price information. The solid line represents the difference of the two data values.

Figure 5: Figure H1: Learning from prices and data value.